MILITARY SPECIFICATION

AMPLIFIER; INTERPHONE, AM-40/AIC (FOR AIRCRAFT)

This specification was approved on the above date by joint action of the Air Force and Navy Departments for use in the procurement of aeronautical supplies, and supersedes the following specification:

NavAer M-659
15 October 1944

This specification consists of this cover sheet and NavAer Specification M-659, dated 15 October 1944, with Amendment No. 2, dated 9 September 1945, without modification.

Copies of this specification may be obtained from the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Custodian:
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NAVY DEPARTMENT
BUREAU OF AERONAUTICS
WASHINGTON, D.C.

NAVAER-M-659
NAVY AERONAUTICAL SPECIFICATION
MANUFACTURER'S SPECIFICATION FOR
INTERPHONE AMPLIFIER AM-40/AIC

15 October 1944

APPROVED: Frank Akers
Captain, U.S.N.
By Direction, Chief of Bureau

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NAVY AERONAUTICAL SPECIFICATION

INTERPHONE AMPLIFIER AM-40/AIC

A. APPLICABLE SPECIFICATIONS

A-1. The following specifications of the issue in effect on date of invitation of bids shall form a part of this specification:

A-1a. Bureau of Ships Specifications:

RE 13A 585 General Requirements for Airborne Electronic Equipments.
RE 13A 825 Type Testing Procedure and Specifications for Airborne Radar and Radio Equipment.

A-1b. Bureau of Aeronautics Drawings:

E-308 Interphone Amplifier AM-40/AIC.

A-1c. Signal Corps - Aircraft Radio Laboratory Drawings:

SC-D-10381-A and associated drawings. Standard Aircraft Radio Aircraft Case & Mounting Base.

A-1d. Specification RE 13A 585 includes a detailed list of applicable specifications.

B. TYPES

B-1. Equipments Covered by This specification - Equipments produced according to these specifications shall be designated Interphone Amplifier AM-40/AIC regardless of manufacturer.

C. MATERIALS AND WORKMANSHIP

C-1. MATERIALS - All materials provided in accordance with this specification shall meet the requirements set forth in specification RE 13A 585 or, if not included in specification RE 13A 585, shall be of the best quality and suitable for the purpose intended.

C-2. WORKMANSHIP - The workmanship on all equipments manufactured in accordance with this specification shall be first class and, in addition, meet requirements set forth in specification RE 13A 585.

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D. GENERAL REQUIREMENTS

D-1. AMPLIFIER UNIT - The AM-40/AIC Interphone Amplifier shall be produced in accordance with the circuit diagram and component parts data on the latest issue of Bureau of Aeronautics drawing E-308.

D-2. TYPE OF CONTROLS - No controls shall be provided for operation by crew members under normal flight conditions. The following controls shall be provided for adjustment of the amplifier characteristics during installation and maintenance or for use by the crew members under abnormal operating conditions:

D-2a. A S.P.S.T. toggle switch (S1 on BuAer drawing E-308) shall be mounted on the face of the amplifier to control the primary power supply. This switch shall be labeled "28 V POWER" with switch positions labeled "ON" and "OFF". The switch handle shall be drilled and a safety wiring stud provided to permit safety wiring in the "ON" position.

D-2b. A S.P.D.T. toggle switch (S2 on BuAer drawing E-308) shall be mounted on the face of the amplifier to permit inserting or removing the altitude switch from the microphone input circuit. This switch shall be labeled "ALTITUDE SWITCH" with the switch positions labeled "IN" and "OUT". The switch handle shall be drilled and safety wiring studs provided to permit wiring in either the "IN" or the "OUT" position.

D-2c. A 10,000 ohm potentiometer (P1 on BuAer drawing E-308) arranged for screwdriver operation only shall be mounted so as to be accessible through the face of the amplifier for use in adjusting the input level of the high impedance input circuit. The access hole for this control shall be provided with a cover which can be safety wired shut after adjustments are completed. This potentiometer shall be labeled "RADIO INPUT LEVEL" to the left of the access hole when facing the amplifier panel, with a curved arrow pointing in the direction of clockwise rotation immediately above the access hole.

D-2d. A 100,000 ohm potentiometer (P2 on BuAer drawing E-308) arranged for screwdriver operation only shall be mounted so as to be accessible through the face of the amplifier for use as a master gain control for the amplifier unit. The access hole for this control shall be provided with a cover which can be safety wired shut after adjustments are completed. This potentiometer shall be labeled "AMPLIFIER GAIN" to the left of the access hole when facing the amplifier panel, with a curved arrow pointing in the direction of clockwise rotation immediately above the access hole.

D-2e. A 4 P.D.T. toggle switch (S3 on BuAer drawing E-308) shall be mounted on the amplifier chassis under the dust cover for use in selecting the average output impedance and amplifier power output which most nearly meets the requirements of any particular installation. This switch shall be labeled "AUDIO OUTPUT POWER" with the switch positions labeled "LO" and "HI" to correspond to low and high power output, respectively, as indicated on

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BuAer drawing E-308. The switch handle shall be drilled and safety wiring studs provided to permit safety wiring in either the "LO" or "HI" position.

D-3. OVERALL AMPLIFIER CHARACTERISTICS - The overall characteristics of the complete AM-40/AIC amplifier shall meet the following requirements under the specified test conditions when the amplifier unit under test is equipped with vacuum tubes and a dynamotor which meet specification RE 13A 585.

D-3a. Maximum Gain from Microphone Input - The maximum gain of each amplifier unit shall be between 29 and 34 db when measured under the following test conditions:

D-3a (1). Adjustment of amplifier controls: -

Control Designation	Control Operated To
S1	"ON"
S2	"OUT"
S3	"HI"
P1	Minimum (Counterclockwise to stop)
P2	Maximum (Clockwise to stop)

D-3a (2) Connections to amplifier receptacle: -

Receptacle Designations	From Pin	To Pin	External Connections
28 volts D.C.	F	G	28 volts D.C.
Relay Control	--	--	None
Mic. Input	C	D	Audio oscillator connected through a blocking condenser. Oscillator adjusted to deliver 1000 cps at 0.5 volt across pins C and D as measured by an accurate high impedance vacuum tube voltmeter (Ballantine or equivalent).
High Imped. In.	--	--	None
Ampl. Out.	A	D	Standard power output meter adjusted to provide a load impedance of 30 ohms.

D-3b. Gain from Microphone Input When Operating Under Reduced Power - The overall gain of each amplifier unit when operating under reduced power shall be between 3 and 6 db less than the gain measured under paragraph D-3a of this specification when measured under the following test conditions:

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D-3b (1) Adjustment of Amplifier Controls - Same as described in paragraph D-3a(1) of this specification except that switch S3 shall be operated to "LO".

D-3b (2) Connections to Amplifier Receptacle - Same as described in paragraph D-3a(2) of this specification except that the power output meter connected to receptacle pins A and D shall be adjusted to provide a load impedance of 120 or 125 ohms.

D-3c. Frequency Response - The frequency response of each amplifier unit shall meet the following requirements when measured under the test conditions described in paragraph D-3c(1) and D-3c (2) of this specification:

Frequency Range	Variation from 1000 c.p.s. Gain
100 c.p.s. and below	At least minus 15 db.
At 300 c.p.s.	Between plus 0 db and minus 6 db.
400 c.p.s. to 3500 c.p.s.	Between plus 1.5 db and minus 3 db following a smooth curve without intermediate resonant points.
At 4000 c.p.s.	Between plus 0 db and minus 7 db.
6000 c.p.s. and above	At least minus 30 db.

D-3c(1) Adjustment of Amplifier Controls - Same as described in paragraph D-3a(1) of this specification.

D-3c(2) Connections to Amplifier Receptacle - Same as described in paragraph D-3a(2) of this specification except that the frequency of the audio oscillator connected to pins C and D shall be adjusted to 100, 300, 500, 2000, 3000, 4000, 6000, and 8000 c.p.s. in addition to the initial 1000 c.p.s. value.

D-3d Distortion - The 1000 cycle distortion of each amplifier unit shall not exceed 9 per cent (exclusive of input oscillator distortion) when measured under the following conditions:

D-3d (1) Adjustment of Amplifier Controls - Same as described in paragraph D-3a(1) of this specification.

D-3d (2) Connections to Amplifier Receptacle: -

Receptacle Designation	From Pin	To Pin	External Connections
28 Volts D.C.	F	G	28 Volts D.C.
Relay Control	-	-	None
Mic. Input	C	D	Audio oscillator connected through a blocking condenser. Oscillator adjusted to deliver 1000 c.p.s. at

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Receptacle Designation	From Pin	To Pin	External Connections
			the voltage required to produce 4 watts output as measured by the power output meter connected to amplifier receptacle pins A and D.
High Imped. In.	-	-	None.
Ampl. Out.	A	D	Standard power output meter adjusted to provide a load impedance of 30 ohms with a bridging type distortion meter of any standard approved manufacture connected across the 30 ohm load.

D-3e. Variation of Power Output with Impedance - The power output of each amplifier unit shall change from 4 watts to a value between 1.5 and 2.0 watts when the load impedance connected to the amplifier output is varied from 30 ohms to 120 or 125 ohms under the following conditions:

D-3e(1) Adjustment of Amplifier Controls - Same as described in paragraph D-3a(1) of this specification.

D-3e(2) Connections to Amplifier Receptacle - Same as described in paragraph D-3d(2) of this specification except that the distortion meter may be omitted and the power output meter connected to amplifier receptacle pins A and D shall be adjusted to 120 or 125 ohms after the input voltage has been adjusted to produce 4 watts into a 30 ohm load.

D-3f Variation of Power Output with Primary Power Voltage - The power output of each amplifier unit shall exceed 2 watts with the amplifier operating from 22 volts d.c. primary power supply under the following conditions:

D-3f(1) Adjustment of Amplifier Controls - Same as described in paragraph D-3a(1) of this specification.

D-3f(2) Connections to Amplifier Receptacle - Same as described in paragraph D-3d(2) of this specification except that the distortion meter may be omitted and the input power voltage connected to amplifier receptacle pins F and G shall be reduced to 22 volts d.c. after the input voltage has been adjusted to produce 4 watts output using standard 28 volt d.c. supply.

D-3g Maximum Dynamotor Load - The maximum current delivered from the 250 volt winding of the dynamotor shall not exceed 55 milliamperes under the following conditions:

D-3g(1) Adjustment of Amplifier Controls - Same as described in paragraph D-3a(1) of this specification.

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D-3g(2) Connections to Amplifier Receptacle - Same as described in paragraph D-3d(2) except that the distortion meter may be omitted.

D-3h. Gain vs. Altitude Switch Positions - The maximum amplifier gain shall vary from the value measured during the test described in paragraph D-3a of this specification in accordance with the following table when measured under the test conditions described in paragraphs D-3h(1) and D-3h(2) of this specification:

Altitude Switch Step	Connections Between Pins On Altitude Switch Socket		Deviation from Maximum Amplifier Gain (Par. D-3a)
	From	To	
1	Pin 5	Pin 7	Minus 2.0 to 4.0 db.
2	Pin 5	Pin 8	Plus 1.0 to 3.0 db.
3	Pin 5	Pin 1	Plus 6.0 to 8.0 db.
4	Pin 5	Pin 2	Plus 10.5 to 13.5 db.

D-3h(1) Adjustment of Amplifier Controls: -

Control Designation	Control Operated To
S1	"ON"
S2	"IN"
S3	"HI"
P1	Minimum (Counterclockwise to stop)
P2	Maximum (Clockwise to stop)

D-3h(2) Connections to Amplifier Receptacle: -

Receptacle Designation	From Pin	To Pin	External Connections
28 Volts D.C.	F	G	28 Volts D.C.
Relay Control	-	-	None
Mic. Input	C	D	Audio oscillator connected through a blocking condenser. Oscillator adjusted to deliver 1000 c.p.s. at any voltage which produces a power output between 1 and 2 watts as measured by the power output meter connected to pins A and D.

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Receptacle Designation	From Pin	To Pin	External Connections
High Imped. In.	-	-	None.
Ampl. Out.	A	D	Standard power output meter adjusted to provide a load impedance of 30 ohms.

D-3i. Gain From High Impedance Input - The maximum gain from the high impedance input shall equal or exceed the gain measured from the microphone input during the tests described in paragraph D-3a of this specification when measured under the following conditions:

D-3i(1) Adjustment of Amplifier Controls: -

Control Designation	Control Operated To
S1	"ON"
S2	"IN" or "OUT"
S3	"HI"
P1	Maximum (Clockwise to stop).
P2	Maximum (Clockwise to stop).

D-3i(2) Connections to Amplifier Receptacle: -

Receptacle Designation	From Pin	To Pin	External Connections
28 Volts D.C.	F	G	28 Volts D.C.
Relay Control	-	-	None.
Mic. Input	-	-	None.
High Imped. In.	B	D	Audio oscillator terminated in nominal output impedance with pins B and D connected across the terminating resistance. Oscillator adjusted to deliver 1000 c.p.s. at any voltage which produces a power output between 1 and 2 watts as measured by the power output meter connected to pins A and D.
Ampl. Out.	A	D	Standard power output meter adjusted to provide a load impedance of 30 ohms.

D-3j. Residual Noise - The residual noise shall not exceed 0.4 milliwatt when measured under the following conditions:

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D-3j(1) Adjustment of Amplifier Controls - Same as described in paragraph D-3a(1) of this specification.

D-3j(2) Connections to Amplifier Receptacle - Same as described in paragraph D-3a(2) of this specification except that no connection shall be made to the microphone input (pins C and D).

D-4. COMPONENTS USED IN ASSEMBLY - All components and small parts used in the manufacture of the Interphone Amplifier AM-40/AIC shall meet the requirements set forth in specification RE 13A 585 except as follows:

D-4a. Temperature Requirements - The completed assembly and all component parts shall meet the service conditions as set forth in paragraph D-2 of specification RE 13A 585 except that the operating and non-operating temperature requirements shall be from minus 40° C to plus 60° C.

D-4b. Relays - All relays provided as components shall be designed for normal operation on 28 volt d.c. and shall meet the following operating limits:

D-4b(1) Shall operate firmly to the armature stop on 19 volts d.c.

D-4b(2) After operation on 28 volts d . with a 10 second soak shall not release until voltage across the winding is reduced to 13 volts.

D-4b(3) When connected to 31 volts d.c. for a period of 4 hours in an ambient temperature of plus 60° C., the temperature rise in the armature winding shall not exceed the accepted safe limits for the type of insulating materials used.

D-4b(4) When in the non-operated condition, all normally closed contacts shall remain closed and there shall be no noticeable motion of the relay armature when viewed with a standard stroboscope while the relay is vibrated in any direction at amplitudes as great as 0.03 inches (0.06 total excursion) at any frequency between 10 to 55 cycles per second.

D-4b(5) When operated on 19 volts d.c., the operated contact shall remain closed and the armature shall remain firmly sealed to the armature stop, under the conditions of vibration described in paragraph D-4b(4), above.

D-4c. Switches - All switches shall be equipped with self-cleaning contacts and be designed and lubricated for operation in temperatures between minus 40° C. and plus 60° C. All insulating materials used shall be molded or laminated bakelite which meets the requirements set forth in specification RE 13A 585.

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D-5 MOUNTING AND WIRING OF COMPONENTS - All components which make up the Interphone Amplifier AM-40/AIC shall be fastened securely to the subpanel assembly, mounted in approved sockets which are fastened to the subpanel assembly, or mounted on approved terminal strips which are fastened to the subpanel assembly. Location, mounting and wiring of all components shall be arranged to permit removing and replacing any single component part without removing the connections to other component parts. If it is necessary to move other component parts to gain access to a particular unit, the wiring to the components which have to be moved shall be of such length and flexibility as to permit the necessary move without disconnecting any of the wiring to the moved parts. Under no circumstances shall small components such as condensers, resistors, etc., be mounted on the terminals of transformers, inductors, vacuum tube sockets or other components which may have to be removed for replacement. For the purpose of this specification, the 4000 cycle low pass filter shall be considered as a single component and designed for removal as a complete unit.

D-6. MOUNTING OF COMPLETE AMPLIFIER - The complete amplifier, associated dynamotor and altitude switch (or space as specified therefore) shall be mounted in an Al-A size Standard Aircraft Radio Case and Mount (Group I on Signal Corps Drawing SC-D-10381-A and associated drawings.)

D-7. DRAWINGS AND INSTRUCTION BOOKS - All drawings and instruction books shall be prepared and submitted in accordance with the requirements set forth in specification RE 13A 585 except as follows:

D-7a. All references to the "Bureau of Ships" in specification relating to drawings (except nameplate drawings) and instruction books shall be changed to read "Bureau of Aeronautics, Radio and Electrical Branch."

D-7b. All drawings and instruction books prepared in accordance to this specification shall be classified as "Restricted".

D-8. MATERIAL TO BE SUPPLIED - The manufacturer shall supply all equipments complete and ready for use including vacuum tubes and one plug with cable clamp of the correct type to fit each specified receptacle. Cables or other external wiring shall not be provided as a component of this equipment.

D-9. PRIMARY POWER REQUIREMENTS - The primary power supply shall not exceed 1.8 amperes at 28 volts d.c. with amplifier operating at maximum power output and including normal microphone current.

E. DETAIL REQUIREMENTS

E-1. APPLICABLE DRAWING - All references to components designations in this specification refer to the designations indicated on Bureau of Aeronautic drawing E-308.

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E-2. CONNECTIONS TO AMPLIFIER - All electrical connections to the amplifier shall be made through a single AN-3102-16s-1P receptacle mounted in the lower left hand corner of the amplifier front panel as seen when facing the amplifier.

E-3. ALTITUDE SWITCH CONTROL - Control of the amplifier gain to compensate for the loss in efficiency of the other components in the intercommunicating system at high altitudes is accomplished by means of a pressure operated rotary switch, or the equivalent, known in general terms as the "Altitude Switch" which is wired to the voltage divider circuit made up of resistors R1 to R5, inclusive, and is cut into or out of the amplifier circuit by toggle switch S2. Unless otherwise specified in the contract, an altitude switch shall not be provided as a component of this amplifier but a space 2-1/4 inches by 2-1/4 inches shall be reserved on the amplifier chassis for future installation of a switch. This space shall be clear of all other apparatus above the chassis within the limits of the dust cover. In the center of this reserved space, an octal base vacuum tube socket shall be mounted and wired to the microphone input voltage divider as shown on Bureau of Aeronautics drawing E-308.

E-4. 4000 CYCLE LOW PASS FILTER - The 4000 cycle low pass filter unit shall be assembled in a case, the dimensions of which shall not exceed 1-1/2 inches by 1-1/2 inches by 4 inches high. All components shall be mounted on a small panel or by another approved method, completely wired and vacuum impregnated as a complete assembly before insertion in an approved metallic case. The completed unit shall meet all of the requirements for Audio Frequency and Power Inductors set forth in specification RE 13A 585.

E-5. ELECTROLYTIC CONDENSERS - Electrolytic condensers C4 to C7, inclusive, shall be designed for a minimum capacitance as specified on Bureau of Aeronautics drawing E-308 when operating at minus 40° C. Each electrolytic condenser shall be arranged for mounting vertically in a standard manner on the subpanel assembly and provided with one solder lug terminal for each positive condenser connection and a single solder lug terminal for all negative (ground) condenser connections. The complete assembly shall meet all requirements as set forth in specification RE 13A 585.

E-6. FUSE - The primary power fuse shall be mounted on the subpanel assembly in accordance with the requirements set forth in specification RE 13A 585. One spare fuse shall be securely fastened to the amplfier dust cover.

E-7. FILTER INDUCTANCES - Filter inductances L1, L2 and L3 shall meet all of the requirements set forth in specification RE 13A 585 and, in addition, shall meet the following special requirements:

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E-7a. Microphone Filter Inductance L1 - The working inductance with 60 milliamperes direct current flowing through the winding shall be between 3 and 5 henries at 1000 c.p.s. The d.c. resistance of the winding shall be between 260 and 275 ohms. The winding conductor shall have a minimum cross section of 25 circular mils.

E-7b. Plate Circuit Filter Inductance L2 - The working inductance with 55 milliamperes direct current flowing through the winding shall be between 0.7 and 1.0 henries at 1000 c.p.s. The d.c. resistance of the winding shall be between 30 and 40 ohms. The winding conductor shall have a minimum cross section of 50 circular mils.

E-7c. Radio Frequency Inductance L3 - The working inductance with 1.8 amperes direct current flowing through the winding shall be between 17 and 20 microhenries at 1000 c.p.s. The d.c. resistance of the winding shall not exceed 0.1 ohms. The winding conductor shall have a minimum cross section of 1600 circular mils. Iron or other magnetic materials shall not be inserted into the magnetic paths of this inductance.

E-8. POTENTIOMETERS - Potentiometers P1 and P2 shall be the electrical and mechanical equivalents of the Navy types specified below except that units of different manufacture which have the same characteristics but slightly different physical size or internal arrangement may be submitted for specific approval:

Potentiometer	Required Navy Type Number
P1	CBZ-631283-20
P2	CBZ-631299-20

E-9. RESISTORS - All resistors shall meet the requirements set forth in specification RE 13A 586 and, in addition, shall be tested by the manufacturer to insure that the actual resistance is within the specified tolerances as indicated for each resistor on Bureau of Aeronautics drawing E-308.

E-10. OUTPUT TRANSFORMER - The output transformer shall meet all the requirements for audio transformers set forth in specification RE 13A 585 and, in addition, shall meet the following special requirements:

E-10a. Impedance Ratio - The transformer shall be designed to match a plate load of 5000 ohms to a line load of 120 ohms when 55 milliamperes are flowing in the plate circuit. The secondary (120 ohm) winding shall be divided into two sections to permit parallel operation into a 30 ohm line.

E-10b. Power Output - The transformer shall be designed to safely deliver 5.5 watts audio power into a 30 ohm line at any frequency from 350 to 4000 cycles per second continuously for 8 hours. The transformer shall also be designed to safely deliver 3.5 watts of audio power into a 120 ohm line at any frequency from 350 to 4000 cycles per second continuously for 8 hours.

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E-10c. Conductor Size - The transformer winding conductors shall have minimum cross sections as specified below by reference to terminal designations indicated on Bureau of Aeronautics drawing E-308:

Winding Designation	From Term.	To Term.	Minimum Cross Section in Circular Mils.
Primary	P	B	60
Secondary	1	2	200
Secondary	3	4	200

E-10d. Frequency Response - The frequency response shall be adequate to permit the complete amplifier to meet the overall frequency requirements.

E-10e. Physical Size and Weight - The physical size and weight of the output transformer shall be kept to the minimum practicable within the limits of the specifications included herein.

E-11. DRAWINGS ATTACHED TO EQUIPMENTS - A black on white schematic drawing showing all wiring connections, actual values of resistance, capacity, etc., for all components, power ratings and other pertinent data required to service the equipments shall be cemented securely to the rear exterior of the subpanel chassis of each equipment. These drawings shall be fully legible and as large as is practicable to mount in the available space. The full designation of the amplifier shall be displayed in large characters on the drawing.

E-12. NAMEPLATES - Nameplates in accordance with specification RE 13A 585 shall be securely attached to or etched into the front panel of each equipment supplied in accordance with this specification.

E-13. SPARE PARTS - Operating and bulk spare parts shall be provided strictly in accordance with the requirements set forth in specification RE 13A 585.

E-14. DYNAMOTOR MOUNTING - Four studs and a 3-conductor plug of the correct type to properly mount a DY-2/ARR-2 dynamotor shall be provided on the amplifier chassis.

E-14a. The four studs on which the dynamotor unit mounts shall be insulated from the chassis by means of gum rubber grommets or the equivalent to provide vibration insulation between the dynamotor and the chassis. These grommets shall be compressed to the point where the dynamotor moves with the chassis as a unit and shows no signs of developing a separate resonant period when the complete assembly is subjected to standard vibration tests.

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E-14b. The three conductor chassis plug for the dynamotor shall be insulated with non-hydroscopic Micalex or similar material which will not carbonize when exposed to momentary flashovers.

E-15. TROPICALIZATION - In addition to other requirements set forth herein, all components and wiring provided as a part of the equipments covered by this specification shall be completely tropicalized using the latest Bureau of Aeronautics approved methods and fungus-resistant lacquers.

E-15a. The manufacturer shall submit a complete description of proposed tropicalization methods as applied to each component and wiring harness for specific Bureau of Aeronautics approval. Tropicalization requirements shall be considered as properly met only when such specific approval of methods has been obtained from the Bureau of Aeronautics.

F. METHODS OF INSPECTION AND TESTS

F-1. All inspections and tests shall conform with the requirements set forth in specification RE 13A 625A except as follows:

F-1a. All references to the "Bureau of Ships" in specification RE 13A 826A shall be changed to "Bureau of Aeronautics, Engineering Division, Radio and Electrical Branch".

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G-1. All packaging, packing and marking for shipment shall be in accordance with requirements set forth in specification RE 13A 585.

H. NOTES

H-1. COGNIZANCE - When equipment in accordance with this specification is purchased by the Navy Department for use in Naval Aircraft the following arrangements will ordinarily apply:

H-1a. Service Requirements - Requirements as to types, quantities and delivery dates are determined by the Bureau of Aeronautics.

H-1b. Procurement - Procurement is initiated by the Bureau of Aeronautics.

H-1c. Contractual, Legal and Financial Matters - Contractual, legal and financial matters are under the cognizance of the Bureau of Supplies and Accounts.

H-1d. Technical Matters - Technical matters are under the cognizance of the Bureau of Aeronautics.

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H-1d(1). Specifications - Specifications are under the cognizance of the Bureau of Aeronautics.

H-1d(2). Inspection - Inspection is the responsibility of the cognizant Inspector or Resident Inspector of Naval Material who represents the Bureau of Aeronautics in such matters. In exceptional cases, inspection may be carried out by representatives of other government establishments.

H-2. NOTICE - Statement on notice shall be as set forth in paragraph I-2 of specification RE 13A 585.

H-3. COPIES -

H-3a. Copies of this specification and drawings and other Navy aeronautical specifications and standard drawings may be obtained from the Navy Department, Bureau of Aeronautics, Washington 25, D. C. Naval activities shall obtain copies from the Naval Aircraft Modification Unit, Johnsville, Pennsylvania (Attn: Records Section, Engineering Division). When requesting specifications, refer to both title and number.

H-3b. Copies of Bureau of Ships specifications may be obtained from the Navy Department, Bureau of Ships, Radio Division, Washington 25, D. C.

H-3c. Manufacturers may obtain specifications from the cognizant Inspector of Naval Material.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15410

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